RADIO FREQUENCY BEACON
AN/ART-(*)

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope. This specification establishes the general design and performance requirements for a radio frequency beacon intended for use in identifying and/or relocating a recoverable aerial target. The beacon shall be capable of operating in conjunction with existing Naval radio direction finding equipment as found in ground stations, chase aircraft, and recovery vehicles.

1.2 Classification.

The radio frequency beacon shall consist of the following:

Item	Applicable Paragraph
Beacon, Radio Frequency	3.5.1.3

1.3 Associated Equipment. The associated equipment not supplied with this equipment consists of the following items:

One (1) antenna type AT- (*)
One (1) set of special support equipment listed under notes (Section 6.5)
One (1) set of interconnecting cables

2. APPLICABLE DOCUMENTS

2.1 General. The following documents of the issue in effect on the date of the invitation for bids, form a part of this specification to the extent specified herein:

Specifications

Military

MIL-E-5400	Electronic Equipment, Aircraft, General Specification For.
MIL-T-5422	Testing, Environmental, Aircraft Electrical Equipment.

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MIL-E-17555	Electronic and Electrical Equipment, Accessories, and Repair Parts: Packaging and Packing Of.
MIL-T-18303	Test Procedures, Production and Inspection, For Aircraft Electronic Equipment, Format For.
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment.

Standards

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment.
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement Of.
MIL-STD-781	Reliability Tests, Exponential Distribution.
MIL-STD-810	Environmental Test Methods.
MIL-STD-831	Test Reports, Preparation Of.

Publications

*IRIG-106-71	Telemetry Standards

(When requesting specifications, drawings, and publications, refer to both title and number. Copies of applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Penna. 19120).

* The IRIG publication may be procured from: Secretariat, Range Commanders Council, White Sands Missile Range, New Mexico, 88002.

3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

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3.2 Parts and Materials. In selection of parts and material, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

(1) Parts and Material as approved by MIL-E-5400 shall be given first consideration

(2) When the characteristics of standard Mil parts and material are such that their use will not fulfill the requirements because of weight, size, performance or other reasons, material and parts shall be used which most nearly meet or exceed requirements of the respective standard parts and material specification.

(3) When a nonstandard part is used, the meeting of general material requirements such as nonflammable, fungus resistant, nontoxic, etc., and the meeting of environmental conditions as required by specification MIL-E-5400 or MIL-T-5422 shall be of prime consideration.

3.2.1 Nonstandard Parts and Material Approval. Approval for use of nonstandard parts and material shall be as outlined in Specification MIL-E-5400 and/or as stipulated in the contract by the procuring agency.

3.3 Design and Construction.

3.3.1 Design Data. No data is required by this specification or by applicable documents referenced in paragraph 2 unless specified in the contract.

3.3.1.1 General. The equipment shall conform to all the applicable requirements of MIL-E-5400 for design, construction, and workmanship, unless otherwise specified herein.

3.3.2 Total Weight. The total weight of the equipment, excluding cables, shall be held to a minimum consistent with good design, and shall not exceed a total weight of six (6) ounces.

3.3.3 Operating Life. The equipment shall operate for 500 hours, continuously or intermittently, without the necessity for repair or readjustment, and within the performance parameters specified in paragraph 3.4.

3.3.4 Cabling and Connections.

3.3.4.1 Cables and Connectors. The equipment shall provide for the use of cables and connectors in accordance with MIL-E-5400. Plugs shall be provided to mate with all external receptacles.

3.3.4.2 Input Power Connection. Input power connections shall be in accordance with paragraph 3.5.1.5.

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3.3.5 Interchangeability. The equipment shall meet the interchangeability requirements as defined in MIL-E-5400

3.3.6 Interference Control. The generation of radio interference shall be controlled within the limits of MIL-STD-461 and IRIG-106-71 Telemetry Standards.

3.3.7 Nomenclature and Nameplates. Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with MIL-N-18307.

3.3.7.1 Type Deisgnation. The parenthesis (*), when used in type designation, shall be replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature. The complete type number shall be used on nameplates, shipping records, and instructions books as applicable.

3.3.8 Standard Conditions. The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests:

Vibration: None
Temperature: Room Ambient (30° ± 10° C)
Altitude: Normal, Ground
Humidity: Room Ambient Up to 90% Relative
Input Power Voltage; 27.5 vdc ± 0.5 vdc

3.3.9 Service Conditions. The equipment shall operate satisfactorily under and after any of the environmental service conditions or reasonable combinations as specified in MIL-E-5400 for Class 2 equipment, except as modified herein.

3.3.9.1 Vibration. The equipment shall operate satisfactorily after being subjected to the vibration requirement of 20 G from 20 to 2,000 Hz in any axis.

3.3.9.2 Shock. The equipment shall operate satisfactorily after being subjected to a half sine wave shock of 50 G for eleven milliseconds duration along each of the three major axes.

3.3.9.3 Acceleration. The equipment shall operate satisfactorily after being subjected to a linear acceleration of 50 G along each of the three major axes.

3.3.9.4 Temperature. The equipment shall operate satisfactorily when subjected to the temperature requirements of Table I Class II specification of MIL-E-5400 except Column I, Continuous Operation, which shall be changed to -30° to +80° C. Column IX, Temperature Extremes, shall apply.

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All other requirements shall be deleted.

3.3.9.5 Altitude. The equipment shall operate satisfactorily after being subjected to an altitude of 70,000 feet.

3.3.9.6 Humidity. The equipment shall operate satisfactorily after exposure to humidity in accordance with MIL-STD-810, Method 507, Procedure I.

3.3.9.7 Leakage (Immersion). The equipment shall operate satisfactorily after submersion in accordance with MIL-STD-810, Method 512, Procedure I.

3.3.9.8 Primary Power Input Requirements. The primary current shall not exceed 350 milliamperes maximum. The beacon shall continue to function with a 28vdc input varying ± 4vdc. Reverse polarity and undervoltage protection shall be provided in the equipment.

3.3.10 Compatibility. The equipment form factor shall be as specified in paragraph 3.5.1.2. The connectors and connections shall be made as specified in paragraph 3.5.1.5.

3.4 Performance.

3.4.1 General. Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under extreme conditions is acceptable, tolerances, or values setting forth acceptable variations from the performance under the standard conditions, shall be as specified in the contract by the procuring activity.

3.4.2 Radio Frequency. The RF beacon shall be designed to transmit on a single frequency within the range of 215 to 260 megahertz, and shall be capable of accepting components tuned to any specific frequency in this range. The equipment shall be delivered tuned to 235.0 megahertz unless otherwise specified by the procuring agency.

3.4.3 Frequency Stability. The RF carrier shall stabilize to ±0.005 percent of that designated by the procuring agency immediately upon application of electrical power.

3.4.4 Beacon Output Impedance. The output impedance shall be 50 ohms nominal over the operating range.

3.4.5 Output Power. The output power of the beacon shall be not less than 1.0 watts peak into a 50 ohm resistive load with a VSWR of 1.5 maximum. A shorted or open circuited signal output to the antenna for a one hour time period shall cause no transmitter damage.

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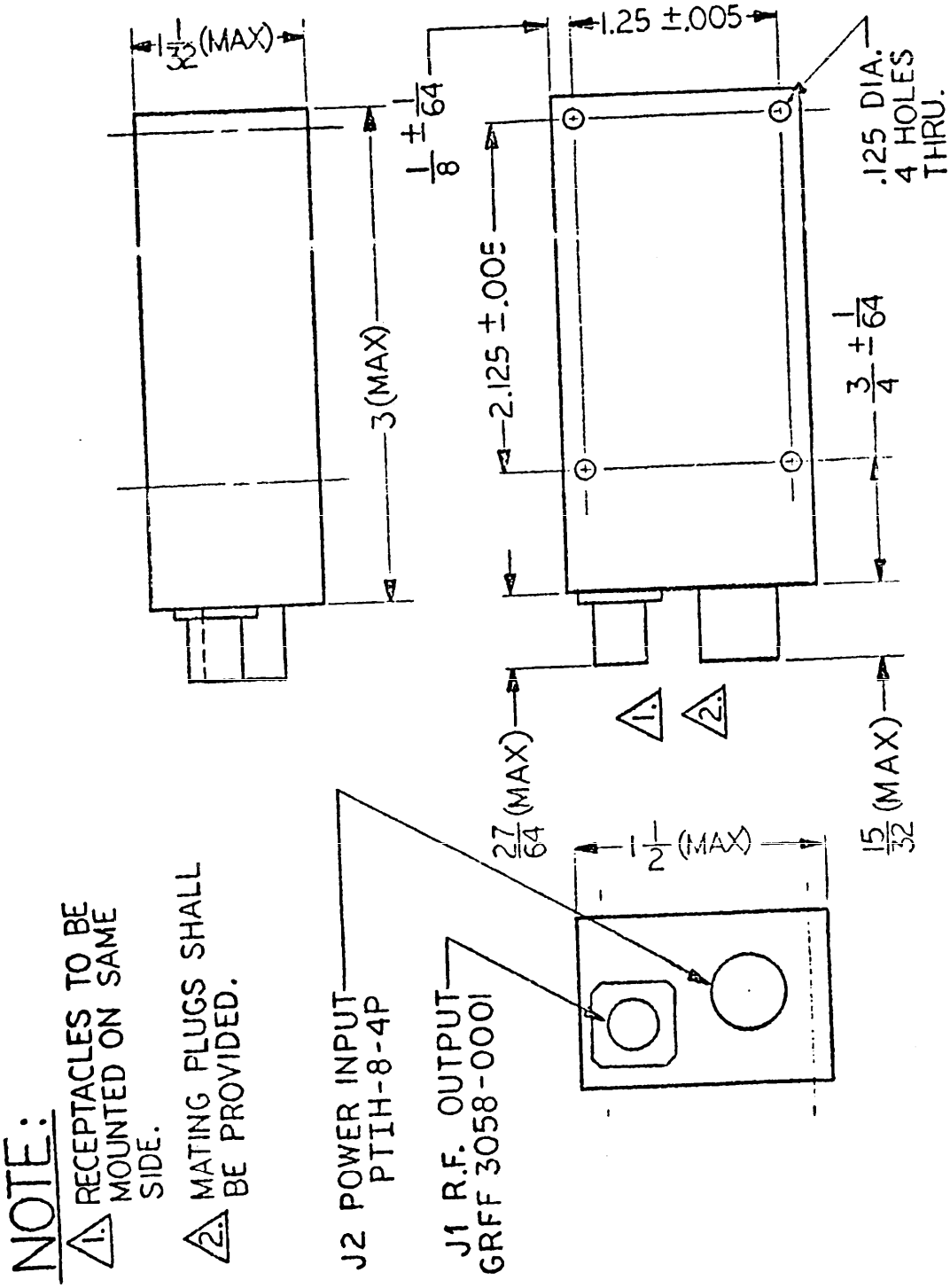


Figure 1. Beacon Transmitter

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3.4.6 Modulation Characteristic. The RF carrier shall be amplitude modulated with a 1,000 ± 20 Hz signal. Carrier modulation shall be within the limits of 30 to 60 percent.

3.4.7 Spurious Emission Limits. Spurious or harmonic emissions shall comply with MIL-STD-461, MIL-STD-462, and spurious emissions as stipulated in IRIG-106-71 over the environmental range.

3.4.8 Modulation Distortion. Distortion of the modulating signal shall not exceed 10 percent maximum over the operating range. Distortion shall be held to the minimum amount consistent with attaining a 1.0 watt peak RF power output.

3.5 Detail Requirements.

3.5.1 Beacon, Radio Frequency. The beacon shall meet the following requirements:

3.5.1.1 Function. The beacon shall generate an amplitude modulated CW carrier at a specific frequency.

3.5.1.2 Form Factor. The beacon form factor shall conform to the dimensions of figure 1.

3.5.1.3 Contents of Unit. The beacon shall contain all the circuitry necessary to generate the CW modulated RF carrier.

3.5.1.4 Controls. No external adjustments shall be required on the beacon.

3.5.1.5 Electrical Connections. Electrical connections shall be provided as follows:

Reference Designation	Receptacle Type	Function
J1	GRFF 3058-0001	RF output
J2	PTIH-8-4P	Power input
	Pin A	+28vdc
	Pin B	Ground
	Pin C & D	No connection

3.6 Reliability. The production equipment shall have a 500 hour minimum mean time between failure (MBTF) when operated in accordance with the performance requirements specified by MIL-STD-781; test level F; or as specified in the contract by the procuring agency.

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4. QUALITY ASSURANCE PROVISION

4.1 Responsibility for Inspection. Unless otherwise specified in the contract, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the government. The government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of Inspections. Inspection of the equipment shall be classified as follows:

(1) Preproduction (First Article) Tests (See 4.3)
(2) Quality conformance inspection (see 4.4)
(3) Inspection of preparation of delivery (see 4.5)

4.3 Preproduction (First Article) Tests - Preproduction tests shall be conducted by the contractor on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the approved test procedure of 4.6. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a government representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the government.

4.3.1 Preproduction (First Article) Test Data - The contractor shall submit all data collected in conducting these tests to the procuring activity for review.

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4.3.2 Scope of Tests. Tests shall consist of such tests as approved by the procuring activity. The contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests. Test methods shall be in accordance with Paragraph 4.6.

4.4 Quality Conformance Inspection. Quality conformance inspection shall consist of the following:

(a) Individual Inspections (see 4.4.1)
(b) Lot inspections (see 4.4.2)

4.4.1 Individual Inspections. Each beacon shall be subjected to the following inspections or tests:

(a) Examination of product (see 4.6.1)
(b) Operational test (see 4.6.2)

These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy, and reliability.

4.4.2 Lot Acceptance Inspections. Lot acceptance inspections may be conducted by the government on beacons which have successfully passed the individual tests. Equipment shall be selected for lot acceptance inspection by the government and shipped on government bills of lading to a designated government activity for tests. The lot sampled for acceptance inspection will be placed in a secure storage area by the contractor at his facility pending determination on lot acceptance. The lot sampled will not be packaged for shipment until a determination on lot acceptance is made or after the expiration of 30 days from the date the sample is taken.

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4.4.2.1 Lot Information. The lot size, sample size and approved quality list will be as specified in the procurement contract.

4.4.2.2 Scope of Tests. As a minimum, each equipment selected for a lot acceptance inspection shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, production, and special tests.

(3) Reliability assurance test specified in 4.6.5.

4.4.3 Test procedures. The procedures used for conducting quality conformance inspection shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the Government Inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.4.4 Equipment Failure. Should a failure occur during any of the quality conformance inspections, the following action shall be taken:

(a) Determine the cause of failure.

(b) Determine if the failure is a material defect or a design defect.

(c) Submit to the procuring activity for approval, proposed corrective action to reduce the possibility of the same failure(s) occurring in future tests.

(d) Where practical, include a test in the individual test to check all equipment for this condition until reasonable assurance is obtained that the defect has been satisfactorily corrected.

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4.5 Inspection of Preparation for Delivery. Sample packages shall be selected in accordance with MIL-STD-105, Inspection Level S-3, AQL (Assured Quality Level) 1.0 percent, and inspected for preservation, packaging, packing, and marking for shipment and storage in accordance with the requirements of Section 5.

4.6 Test Methods.

4.6.1 Examination of Product. Each equipment shall be carefully examined to determine that the material and workmanship requirements have been met.

4.6.2 Operational Test. Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.6.3 Environmental Tests. Environmental tests shall be in accord with MIL-T-5422 or MIL-STD-810.

4.6.4 Interference Tests. Interference tests shall be in accord with MIL-STD-461 and MIL-STD-462.

4.6.5 Reliability Test. The reliability test shall be in accordance with MIL-STD-781, Test Plan XXIX, Test Level F, except as specified herein. Three (3) equipments shall be subjected to this test. Each equipment shall be operated for 200 hours and the total accumulative time shall be 600 hours. The length of the heating period shall be (1) one hour. The equipment shall be on during the heating period and off during the cooling period. Measurement of performance (see 3.4) shall be made at least once during each heating period. The test shall be considered completed satisfactorily if no relevant failures are experienced during the 600 hours cumulative time. Equipment subjected to this test may be restored and delivered as items under the contract provided they pass quality conformance inspection (see 4.4). A record of the corrective maintenance for each failure experienced during this test shall be maintained.

5. PREPARATION FOR DELIVERY

5.1 General. The equipment shall be preserved, packaged, and marked for the level of shipment specified in the contract in accordance with MIL-E-17555.

6. NOTES

6.1 Intended Use. The equipment herein will be installed in an airborne target.

6.2 Performance Objective. Minimum size, and weight, simplicity of operating, ease of maintenance, and an improvement in the performance

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and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts, and process other than those specified in specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring agency for consideration. Each request shall be accompanied by complete supporting information.

6.3 Ordering Data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Levels of packaging and packing of shipments.
(c) Lot size.
(d) Sample size and assured quality level.

6.4 Test Values. Normal and limiting values of performance data shall be determined at an input of 28 vdc $\pm$ 4.0 vdc. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standards of performance.

6.5 Special Support Equipment. The special support equipment or its equivalent recommended to ascertain beacon performance is listed as follows:

(a) Regulated power supply, Power Design Inc. Model 5015.

(b) Power output meter, Bird Corporation, Thruline Model 43.

(c) Fixed load, 50 ohms, Bird Termaline Coaxial Resistor, Model 80M.

(d) RF detector, PRD Model 613M.

(e) Oscilloscope, Tektronix Model 150A and associated Model 151A amplifier.

(f) Adjustable attenuator, General Radio, GR 874-GAL.

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DOCUMENT IDENTIFIER AND TITLE

MIL-R-81304A(AS) RADIO FREQUENCY BEACON AN/ART-(*)

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

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